

Supplementary Materials for

Finite Prevention Windows for HIV Post-Exposure Prophylaxis: Irreversible Proviral Integration Defines Route-Specific Intervention Limits

A.C. Demidont*

Nyx Dynamics LLC, Philadelphia, PA 19107, USA

*Correspondence: ac.demidont@nyxdynamics.com

ORCID: 0000-0002-9216-8569

This PDF file includes:

- Sections S1–S6: Mathematical development with complete proofs (unchanged from v1)
- Section S7: Multiscale within-host model implementation
- Section S8: NHP concordance — per-timepoint detail
- Section S9: Parameter tables and stochastic VL methods
- Section S10: City-stratified extended methods
- Section S11: Figure inventory

Note on v2 changes. The mathematical content in Sections S1–S6 (theorems, proofs, lemmas, propositions) is unchanged from v1. Numerical worked examples in Section S6 have been updated to reflect verified values from the multiscale model implementation (commit 37e27ea). New material in Sections S7–S8 documents the multiscale implementation and NHP concordance.

S1 Probability Space, State Definitions, and Assumptions

Consider a single HIV exposure event e occurring at time $t = 0$ on probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $\{\mathcal{F}_t\}_{t \geq 0}$.

S1.1 Within-host states

Definition S1.1 (Within-host infection states). $Z(t) \in \{0, 1, 2\}$:

- $Z(t) = 0$ (Susceptible): no productive infection; $I(t) < I^*$.
- $Z(t) = 1$ (Seeded): replicative focus established ($I(t) \geq I^*$) but reservoir below irreversibility threshold ($R(t) < R^*$).
- $Z(t) = 2$ (Integrated): $R(t) \geq R^*$. Absorbing state under Assumption A1.

Definition S1.2 (Transition stopping times).

$$T_{\text{seed}} = \inf\{t \geq 0 : Z(t) \geq 1\}, \quad T_{\text{int}} = \inf\{t \geq 0 : Z(t) = 2\}.$$

We assume $T_{\text{seed}} \leq T_{\text{int}}$ almost surely.

Definition S1.3 (Cumulative transition probabilities). $P_{\text{seed}}(t) = \mathbb{P}(T_{\text{seed}} \leq t)$, $P_{\text{int}}(t) = \mathbb{P}(T_{\text{int}} \leq t)$. Since $T_{\text{seed}} \leq T_{\text{int}}$ a.s., $P_{\text{seed}}(t) \geq P_{\text{int}}(t)$ for all $t \geq 0$.

S1.2 Assumptions

Assumption S1.1 (Irreversibility of integration). Once $Z(t) = 2$, the state is absorbing: $\mathbb{P}(Z(s) = 0 \mid Z(t) = 2) = 0$ for all $s > t$.

Biological grounding: Integrated provirus persists for the lifetime of the host cell and propagates to daughter cells via clonal expansion. ART suppresses replication ($I \rightarrow 0$) but $dR/dt = \alpha I \rightarrow 0$: the reservoir is static under ART but does not shrink. No approved therapy achieves eradication.

Assumption S1.2 (Stage-dependent drug efficacy).

$$\varepsilon(Z(t)) = \begin{cases} \varepsilon_{\max} & Z(t) = 0, \\ \varepsilon_{\text{mid}} & Z(t) = 1, \\ \varepsilon_{\min} = 0 & Z(t) = 2, \end{cases} \quad \text{with } 1 \geq \varepsilon_{\max} \geq \varepsilon_{\text{mid}} \geq 0.$$

Assumption S1.3 (Monotonicity of integration). $P_{\text{int}}(t)$ is non-decreasing, $P_{\text{int}}(0) = 0$, $\lim_{t \rightarrow \infty} P_{\text{int}}(t) = 1$.

Assumption S1.4 (Transmission-competent exposure). Source plasma viral load is at or above the $U=U$ threshold (≥ 200 copies/mL sustained). The framework is silent on exposures from virologically suppressed sources, for whom transmission risk is functionally zero (Rodger 2019, PARTNER2; Bavinton 2018, Opposites Attract).

S2 Master Equation

Proposition S2.1 (Master equation). *Under Assumption A2,*

$$E_{\text{PEP}}(t) = (1 - P_{\text{seed}}(t))\varepsilon_{\max} + (P_{\text{seed}}(t) - P_{\text{int}}(t))\varepsilon_{\text{mid}} + P_{\text{int}}(t)\varepsilon_{\min} \quad (\text{S1})$$

Proof. Partition the sample space at time t by $Z(t)$:

$$\mathbb{P}(Z = 0) = 1 - P_{\text{seed}}, \quad \mathbb{P}(Z = 1) = P_{\text{seed}} - P_{\text{int}}, \quad \mathbb{P}(Z = 2) = P_{\text{int}}.$$

These events are mutually exclusive and exhaustive. By the law of total expectation:

$$E_{\text{PEP}}(t) = \mathbb{E}[\varepsilon(Z(t))] = (1 - P_{\text{seed}})\varepsilon_{\max} + (P_{\text{seed}} - P_{\text{int}})\varepsilon_{\text{mid}} + P_{\text{int}}\varepsilon_{\min}. \quad \square$$

S3 The Finite Prevention Window Theorem

Theorem S3.1 (Finite Prevention Window). *Under Assumptions A1–A3, $E_{\text{PEP}}(t)$ satisfies:*

- (i) $E_{PEP}(t) \leq (1 - P_{int}(t)) \epsilon_{\max}$;
- (ii) $E_{PEP}(t)$ is non-increasing in t ;
- (iii) $E_{PEP}(t) \rightarrow 0$ as $t \rightarrow \infty$;
- (iv) for any $\eta \in (0, 1)$, $t_{crit}(\eta) = \inf\{t : E_{PEP}(t) < \eta\}$ is finite.

Proof. Part (i) — Upper bound. Replace ϵ_{mid} with $\epsilon_{\max}$ in (S1):

$$E_{PEP}(t) \leq (1 - P_{seed})\epsilon_{\max} + (P_{seed} - P_{int})\epsilon_{\max} + P_{int}\epsilon_{\min} = (1 - P_{int})\epsilon_{\max} + P_{int}\epsilon_{\min}.$$

With $\epsilon_{\min} = 0$: $E_{PEP}(t) \leq (1 - P_{int}(t))\epsilon_{\max}$.

Part (ii) — Monotone decay. For $t_1 < t_2$, let $\Delta_S = P_{seed}(t_2) - P_{seed}(t_1) \geq 0$ and $\Delta_I = P_{int}(t_2) - P_{int}(t_1) \geq 0$. Then

$$E_{PEP}(t_1) - E_{PEP}(t_2) = \Delta_S(\epsilon_{\max} - \epsilon_{mid}) + \Delta_I(\epsilon_{mid} - \epsilon_{\min}) \geq 0.$$

Part (iii) — Convergence. By A3, $P_{int}(t) \rightarrow 1$. Since $T_{seed} \leq T_{int}$ a.s., $P_{seed}(t) \rightarrow 1$. Limits give $E_{PEP}(t) \rightarrow 0 \cdot \epsilon_{\max} + 0 \cdot \epsilon_{mid} + 1 \cdot \epsilon_{\min} = 0$.

Part (iv) — Finite critical time. By (ii), E_{PEP} is non-increasing; by (iii), $E_{PEP} \rightarrow 0 < \eta$. Therefore $\{t : E_{PEP}(t) < \eta\}$ is non-empty, and $t_{crit}(\eta)$ is finite. $\square$

S4 Hazard-Rate Decomposition of Efficacy Loss

Definition S4.1 (Hazard rates). $\lambda_{seed}(t) = f_{seed}(t)/(1 - P_{seed}(t))$, $\lambda_{int}(t) = f_{int}(t)/(P_{seed}(t) - P_{int}(t))$.

Proposition S4.1 (Decomposition). Under Assumptions A1–A3 with absolute continuity,

$$\frac{dE_{PEP}}{dt} = -f_{seed}(t)\Delta\epsilon_1 - f_{int}(t)\Delta\epsilon_2 \leq 0 \tag{S2}$$

where $\Delta\epsilon_1 = \epsilon_{\max} - \epsilon_{mid}$ and $\Delta\epsilon_2 = \epsilon_{mid} - \epsilon_{\min}$.

Proof. Differentiating (S1):

$$\frac{dE_{PEP}}{dt} = -f_{seed}\epsilon_{\max} + (f_{seed} - f_{int})\epsilon_{mid} + f_{int}\epsilon_{\min} = -f_{seed}\Delta\epsilon_1 - f_{int}\Delta\epsilon_2 \leq 0. \quad \square$$

Remark. Equation S2 decomposes efficacy loss into a seeding channel (f_{seed} , gap $\Delta\epsilon_1$) and an integration channel (f_{int} , gap $\Delta\epsilon_2$). At early times, seeding dominates; as mass accumulates in state 1, the integration channel accelerates terminal efficacy collapse.

S5 Route Compression and Inoculum-Monotone Hitting Times

S5.1 ODE system

The standard target-cell-limited within-host model:

$$\frac{dT}{dt} = \lambda - dT - \beta TV, \quad \frac{dI}{dt} = \beta TV - \delta I, \quad \frac{dV}{dt} = pI - cV, \quad \frac{dR}{dt} = \alpha I \quad (\text{S3})$$

where T = target cells, I = infected cells, V = free virions, R = cells with stable proviral integration.

Lemma S5.1 (Absorbing state). *The set $\mathcal{A} = \{R > 0\}$ is positively invariant. If $R(t_0) > 0$ then $R(t) > 0$ for all $t > t_0$.*

Proof. $dR/dt = \alpha I \geq 0$, so R is non-decreasing. Under ART, $I \rightarrow 0$ and $dR/dt \rightarrow 0$: the reservoir is static but does not shrink. $\square$

S5.2 Route-dependent initial conditions

$V^{(m)}(0) = 1$ virion/mL (mucosal, post-epithelial attenuation) versus $V^{(p)}(0) = 10^3$ virions/mL (parenteral, direct intravascular delivery). All ODE parameters in (S3) are identical across routes.

Lemma S5.2 (Inoculum-monotone hitting times). *Solutions of (S3) with $V(0) = v_0 < \tilde{V}(0) = \tilde{v}$ satisfy $\tilde{I}(t) \geq I(t)$ and $\tilde{R}(t) \geq R(t)$ for all $t \geq 0$. Therefore $T_{\text{int}}(\tilde{v}) \leq T_{\text{int}}(v_0)$ a.s.*

Proof. The subsystem (I, V, R) is cooperative for fixed T :

$$\frac{\partial(\beta TV)}{\partial V} \geq 0, \quad \frac{\partial(pI)}{\partial I} \geq 0, \quad \frac{\partial(\alpha I)}{\partial R} = 0.$$

By comparison theorems for cooperative ODE systems[7], solutions are order-preserving in initial conditions. Since $\tilde{V}(0) > V(0)$: $\tilde{V}(t) \geq V(t)$ and $\tilde{I}(t) \geq I(t)$ for all $t \geq 0$. Then

$$\tilde{R}(t) = \alpha \int_0^t \tilde{I}(s) ds \geq \alpha \int_0^t I(s) ds = R(t),$$

so $T_{\text{int}}(\tilde{v}) \leq T_{\text{int}}(v_0)$. $\square$

S5.3 Route compression corollary

Assumption S5.1 (Stochastic dominance). $P_{\text{int}}^{(p)}(t) \geq P_{\text{int}}^{(m)}(t)$ for all $t \geq 0$. Follows from Lemma S5.2 under route-specific inocula.

Corollary S5.1 (Route compression). *Under Assumptions A1–A4:*

- (i) $E_{PEP}^{(p)}(t) \leq E_{PEP}^{(m)}(t)$;

$$(ii) \ t_{crit}^{(p)}(\eta) \leq t_{crit}^{(m)}(\eta).$$

Proof. (i) The weight on $\epsilon_{\max}$ equals $1 - P_{\text{seed}}(t)$, which is smaller for parenteral; the weight on $\epsilon_{\min} = 0$ equals $P_{\text{int}}(t)$, which is larger for parenteral. So $E_{\text{PEP}}^{(p)} \leq E_{\text{PEP}}^{(m)}$.

(ii) Both functions are non-increasing; $t_{crit}^{(p)}(\eta) = \inf\{t : E_{\text{PEP}}^{(p)} < \eta\} \leq \inf\{t : E_{\text{PEP}}^{(m)} < \eta\} = t_{crit}^{(m)}(\eta)$. $\square$

S6 Population-Level PEP Efficacy Bound

Definition S6.1. $\bar{E}_{\text{PEP}} = \mathbb{E}[E_{\text{PEP}}(T_{\text{access}})] = \int_0^\infty E_{\text{PEP}}(t) dF_{\text{access}}(t)$.

Proposition S6.1 (Population-level bound). *Under Assumptions A1–A4,*

$$\bar{E}_{\text{PEP}} \leq F_{\text{access}}(t_{crit}) \epsilon_{\max} + (1 - F_{\text{access}}(t_{crit})) \eta \quad (\text{S4})$$

Proof. Partition the integral at $t_{crit}(\eta)$:

$$\bar{E}_{\text{PEP}} = \int_0^{t_{crit}} E_{\text{PEP}}(t) dF_{\text{access}} + \int_{t_{crit}}^\infty E_{\text{PEP}}(t) dF_{\text{access}} \leq \epsilon_{\max} F_{\text{access}}(t_{crit}) + \eta (1 - F_{\text{access}}(t_{crit})),$$

using $E_{\text{PEP}}(t) \leq \epsilon_{\max}$ for $t \leq t_{crit}$ and $E_{\text{PEP}}(t) < \eta$ for $t > t_{crit}$ (Theorem S3.1(iv)). $\square$

S6.1 Worked example (v2 verified values)

Using the multiscale model output at the canonical operating point ($V_0 = 10^3$ parenteral, $\text{CV} = 0.3$ on β, c, δ, α ; commit 37e27ea), $t_{crit}^{(p)}(0.05) \approx 34.5$ h. Modeling access delay as log-normal (median 96 h, geometric SD 2.0):

$$F_{\text{access}}(34.5 \text{ h}) = \Phi\left(\frac{\ln 34.5 - \ln 96}{\ln 2}\right) = \Phi(-1.476) \approx 0.0699.$$

With $\eta = 0.05$ and $\epsilon_{\max} = 0.95$:

$$\bar{E}_{\text{PEP}} \leq 0.0699 \times 0.95 + 0.9301 \times 0.05 \approx 0.113.$$

Envelope reading: expected population PEP efficacy below approximately 11.3%.

S6.2 Sensitivity of the envelope to F_{access} parameterization

The median time-to-PEP-access for PWID is not directly anchored by primary timing data in the cited references (Taylor 2019[10] is a 3-page commentary; Baugher 2025[11] is NHBS-PWID surveillance focused on PrEP/syringe metrics, not PEP timing). The median is therefore a modeling choice informed by qualitative discussion of structural barriers in those references. Sensitivity:

Table S1: Envelope sensitivity to F_{access} median (GSD = 2.0, $t_{\text{crit}}^{(p)} = 34.5$ h, $\eta = 0.05$, $\varepsilon_{\text{max}} = 0.95$).

Median	$F_{\text{access}}(34.5 \text{ h})$	Envelope $\bar{E}_{\text{PEP}}$
72 h (JID-consistent)	0.144	$\leq 18.0\%$
96 h (canonical)	0.070	$\leq 11.3\%$
110 h	0.047	$\leq 9.2\%$
120 h	0.036	$\leq 8.2\%$

The envelope bound is the most generous (least pessimistic) reading of population-level PEP efficacy under the framework; in independent cascade-based modeling of structural access (companion analysis; see main text Population-level efficacy section), point estimates of $P(R_0 = 0)$ for PWID under current US policy are 0.003–0.7%, one to four orders of magnitude below the envelope. Both lines of evidence support the qualitative claim that PEP cannot serve as a population-level safety net for PWID under current structural conditions; the envelope provides the analytical bound, the cascade evidence provides the central estimates.

S6.3 Limitation: F_{access} denominator

F_{access} derives from population-wide PEP-access surveillance, which includes exposures from both transmission-competent and virologically suppressed sources. Under U=U (Assumption A4), exposures from suppressed sources are outside the framework’s domain. Restricting F_{access} to transmission-competent exposures only would reduce the relevant denominator and yield a tighter (smaller) F_{access} and therefore a tighter envelope. The reported envelope is thus conservative with respect to U=U scoping: the true within-domain envelope is no larger than the reported value.

S7 Multiscale Within-Host Model: Implementation

The multiscale within-host model combines stochastic founder-phase dynamics with deterministic ODE integration once productive infection is established. This section documents the implementation; all code is reproducible from SRC/multiscale_model/ at commit 37e27ea of the project repository.

S7.1 Phase 1: Stochastic founder-phase tau-leaping

At low founder population ($V_0 \leq I_{\text{handoff}} = 10$), the dynamics are sensitive to extinction events; deterministic integration of (S3) systematically underestimates the extinction probability and overestimates the rate of reservoir establishment. Phase 1 uses tau-leaping integration of the discrete-state Markov chain with reactions:

$$T \xrightarrow{\beta TV} I, \quad I \xrightarrow{\delta} \emptyset, \quad I \xrightarrow{p} I + V, \quad V \xrightarrow{c} \emptyset, \quad I \xrightarrow{\alpha} R.$$

A fixed-delay buffer of $\tau_{\text{eclipse}} = 0.9$ days (≈ 22 h, Perelson 1996 Table 2)[5] is enforced between cell infection and integration competence, ensuring that the lower bound on T_{int} matches the biological eclipse

floor regardless of population dynamics.

A realization is classified as *extinct* if both $I(t)$ and $V(t)$ reach zero before the population threshold $I_{\text{handoff}} = 10$ is exceeded. A realization is classified as *integrated* if $R(t) \geq R^* = 10$ cells/mL is reached. For mucosal exposure ($V_0 = 1$), extinction probability under $\text{CV} = 0.3$ is $\approx 68\%$; for parenteral exposure ($V_0 = 10^3$, $\text{CV} = 0.3$), extinction is essentially zero ($< 0.5\%$).

S7.2 Phase 2: Deterministic ODE handoff

Once $I(t)$ exceeds $I_{\text{handoff}} = 10$, the dynamics transition to deterministic integration of (S3) using `scipy.integrate.solve_` with adaptive step size. The handoff continues until $R(t) \geq R^*$, at which point T_{int} is recorded.

S7.3 Heterogeneity

Inter-individual variability is modeled as multiplicative log-normal noise on $\{\beta, c, \delta, \alpha\}$ with coefficient of variation 0.3. The noise parameters are sampled once per realization; deterministic phase parameters use the sampled values throughout. The $\text{CV} = 0.3$ value is consistent with the standard within-host model heterogeneity assumption used in earlier published noise-anchored models.

S7.4 Sample size and seeding

We use $N = 500$ realizations per (route, V_0 , CV) cell. Random seeds are deterministic: `rng = np.random.default_rng(ba` where k is the realization index. This guarantees byte-identical reproducibility from a cloned repository at the cited SHA. Sensitivity to N at the canonical operating point is reported in main text Methods; t_{crit} estimates are stable to ± 0.5 h across $N \in \{200, 500, 1000, 5000\}$.

S7.5 Output: t_{crit} and $P_{\text{int}}(t)$

For each cell, the empirical CDF $P_{\text{int}}(t)$ is constructed from the N -realization integration-time distribution. The route-specific efficacy curve $E_{\text{PEP}}(t)$ is computed from $P_{\text{int}}(t)$ via the master equation (S1), and $t_{\text{crit}}(\eta)$ is the smallest t at which $E_{\text{PEP}}(t) < \eta$. Reported t_{crit} values:

Table S2: Verified t_{crit} at $\eta = 0.05$ from SRC/multiscale_model/results_v3/heterogeneity_summary.csv.

Route	V_0	CV	$t_{\text{crit}}(0.05)$ [h]
Parenteral	10^3	0.0	32.5
Parenteral	10^3	0.3	34.5
Parenteral	10^3	0.5	36.0
Mucosal	1	0.0	57.5
Mucosal	1	0.3	60.5
Mucosal	1	0.5	60.0

The compression ratio at the canonical operating point ($CV = 0.3$) is $60.5/34.5 \approx 1.75\times$.

S8 NHP Concordance — Per-Timepoint Detail

The multiscale model output was compared to NHP empirical protection rates at the timepoints reported in Tsai 1998[8] and Otten 2000[9]. Concordance is defined as $|E_{\text{PEP}}(\text{delay}) - \text{NHP protection}| < 0.20$ (within 20 percentage points).

Table S3: Per-timepoint NHP concordance at the canonical operating point ($CV = 0.3$). Reproducible from SRC/multiscale_model/results_v3/nhp_concordance.csv.

Study	Route	Delay [h]	NHP protection	Model E_{PEP}	diff [pp]	Concordant
Otten 2000	mucosal	12	1.00	0.95	5.0	yes
Otten 2000	mucosal	36	1.00	0.95	5.0	yes
Otten 2000	mucosal	72	0.50	0.016	48.4	no
Tsai 1998 IV	parenteral	24	1.00	0.50	50.0	no
Tsai 1998 IV	parenteral	48	0.50	0.00	50.0	no

Compression-ratio concordance. The model’s mucosal/parenteral t_{crit} ratio at $\eta = 0.05$ is $60.5/34.5 \approx 1.75\times$. The empirical NHP compression ratio between Tsai 1998 IV (parenteral half-protection at 48 h) and Otten 2000 (mucosal half-protection at 72 h) is $72/48 = 1.50\times$. The model’s compression value is concordant with the empirical NHP range of $1.5\text{--}2\times$.

Per-timepoint absolute discordance. The within-host kinetic model under-predicts protection at:

- Tsai 1998 IV at 24 h (model 0.50 vs. NHP 1.00),
- Tsai 1998 IV at 48 h (model 0.00 vs. NHP 0.50),
- Otten 2000 at 72 h (model 0.016 vs. NHP 0.50).

We interpret this gap as evidence of NHP-specific protective contributions not captured by the within-host kinetic framework alone. Possible mechanisms include innate immunity contributions during the eclipse phase, mucosal containment by epithelial defense, and the small NHP cohort size ($n = 2\text{--}4$ per group) producing wider empirical confidence intervals on observed protection rates than the model’s first-passage formulation accommodates. The compression-ratio result and the absolute-timepoint result are independent: the compression result is robust to the model’s domain limitations; absolute-timepoint discordance constrains the interpretation of the model as a kinetic-only ceiling rather than a full mechanistic prediction of NHP protection.

S9 Parameter Tables

S9.1 ODE system parameters (Perelson 1996 anchored)

Table S4: ODE system parameters anchored on Perelson et al. 1996 (PMID 8599114).

Symbol	Description	Value	Units	Source
δ	Infected cell death rate	0.7	day^{-1} ($t_{1/2} \approx 1.6\text{d}$)	Perelson 1996, Table 1
c	Viral clearance rate	23	day^{-1} ($t_{1/2} = 0.24\text{d} \approx 6\text{h}$)	Perelson 1996, Table 1
τ	Viral generation time	2.6	days ($\approx 62\text{h}$)	Perelson 1996, Table 2
Eclipse phase	Min time to integration competence	0.9	days ($\approx 22\text{h}$)	Perelson 1996, Table 2
β	Infection rate constant	2.4×10^{-5}	$\text{mL} \cdot \text{virion}^{-1} \cdot \text{day}^{-1}$	Standard model
p	Virion production per infected cell	10^3	virions/cell/day	Standard model
α	Stable integration rate	10^{-3}	day^{-1}	Calibrated

Parameters c and δ directly from Table 1; eclipse phase and generation time from Table 2. *Note:* the reference $\log_{10} \text{VL} = 4.5$ for PWID is from NHBS/NHAS surveillance, not from Perelson 1996 (whose patients had mean $\log_{10} \text{VL} \approx 5.3$).

S9.2 Drug efficacy parameters

Table S5: Stage-dependent drug efficacy parameters.

Parameter	Value	State $Z(t)$	Justification
$\epsilon_{\max}$ (pre-seeding)	0.95	$Z(t) = 0$	Near-complete suppression of initial RT; consistent with tenofovir/FTC NHP protection data
ϵ_{mid} (seeded/pre-int)	0.50	$Z(t) = 1$	Partial suppression of established replicative focus; sensitivity $\pm 40\%$ shifts $t_{\text{crit}} < 6\text{h}$
$\epsilon_{\min}$ (integrated)	0 (fixed)	$Z(t) = 2$	ART does not eradicate provirus; set to 0 by A2 and Lemma S5.1

S9.3 Route-specific initial conditions and thresholds

Table S6: Route-specific initial conditions and threshold parameters.

Parameter	Mucosal	Parenteral	Note
$V(0)$ virions/mL	1	10^3	3-log difference drives route compression; post-epithelial attenuation vs. direct IV delivery
$T(0)$ CD4 cells/mL	10^6	10^6	Identical; peripheral blood
$I(0)$ and $R(0)$	0	0	No infected cells or reservoir at exposure
I^* (seeding threshold)	100 cells/mL	100 cells/mL	Sensitivity: $[10, 1000]$; t_{crit} shifts $< 8\text{h}$; compression ratio stable
R^* (integration threshold)	10 cells/mL	10 cells/mL	Sensitivity: $[1, 100]$; parenteral/mucosal ratio stable $1.5\text{--}2\times$
I_{handoff} (Phase 1 $\rightarrow$ 2)	10 cells/mL	10 cells/mL	Stochastic-to-deterministic transition threshold

S9.4 Multiscale simulation parameters

Table S7: Multiscale simulation parameters.

Parameter	Value	Description
N Monte Carlo replications	500 / cell	Per (route, V_0 , CV) cell; total $\sim 27,000$ across full sweep
Heterogeneity CV	0.3	Multiplicative log-normal on $\{\beta, c, \delta, \alpha\}$
Random seed	deterministic	<code>rng = np.random.default_rng(42*1,000,000+k)</code>
Tau-leaping step	adaptive	Phase 1, founder-phase
ODE integrator	<code>solve_ivp</code> (LSODA)	Phase 2, deterministic handoff
Eclipse delay buffer	0.9 days ($\approx 22\text{h}$)	Fixed delay between cell infection and integration competence
CI type	95% credible interval	From stochastic T_{int} distribution

S10 Stochastic VL Analysis — Extended Methods

S10.1 Source VL distributions

Expected PEP efficacy was computed as a Monte Carlo integral over source viral load:

$$\mathbb{E}[E_{\text{PEP}}(t)] = \int E_{\text{PEP}}(t, \text{VL}) P(\text{VL}) d\text{VL} \quad (\text{S5})$$

Table S8: Source VL distributions parameterized from surveillance data. All log₁₀-normal.

Distribution	Mean log ₁₀	SD	Empirical basis
PWID untreated/unsuppressed	4.5	1.1	NHBS/NHAS surveillance; chronic untreated HIV-1 in PWID networks
PWID mixed treatment	3.2	1.5	Mixture: suppressed (~ 1.5) + unsuppressed (~ 4.5) in partially treated PWID networks
General HIV+ community	3.8	1.2	Mixed treatment uptake; heterogeneous MSM/general community networks
Acute-infection-enriched	5.0	0.9	High-transmission networks; acute viremia peak; recent-infection clustering

Note: PWID untreated mean log₁₀ = 4.5 is from NHBS/NHAS surveillance, NOT from Perelson 1996.

S10.2 Three-regime temporal classification

Table S9: Three-regime CI-width classification for the PWID untreated distribution.

Regime	Hours (PWID untreat.)	Peak CI width	Clinical interpretation
1 — Early window	0–22.0 h	<15 pp	VL uncertainty low-impact; initiate PEP immediately regardless of source VL
2 — Critical window	22.0–78.4 h (PWID/acute); 22.0–88.2 h (mixed/general)	39.7 pp at 49 h	Maximal VL uncertainty; source VL testing clinically actionable; VL premium peaks at 21 h
3 — Late window	78.4–120 h (PWID/acute); 88.2–120 h (mixed/general)	<10 pp (collapsed)	Efficacy uniformly poor; VL irrelevant; $P(\text{futile}) > 90\%$

The Regime 1/2 boundary at 22.04 h is consistent across all four VL distributions (eclipse phase lower bound, Perelson 1996). The Regime 2/3 boundary is distribution-dependent: 78.37 h for PWID untreated and acute-infection-enriched distributions (higher VL accelerates CI collapse); 88.16 h for mixed-treatment and general-community distributions. This distribution-specificity is biologically coherent: higher inoculum populations exhaust the critical window earlier.

S10.3 VL knowledge premium and non-monotonic plateau

VL knowledge premium = absolute difference in expected $E_{\text{PEP}}(t)$ between known vs. unknown source VL. Peak values: 3.76 pp at 21 h (acute-enriched); 2.94 pp (PWID untreated); 1.87 pp (general community); 1.27 pp (mixed Tx). All distributions peak at 21 h — maximal VL-knowledge value at critical window onset.

S11 City-Stratified Analysis — Extended Methods

S11.1 Data source

City-level HIV surveillance data were extracted from AIDSVu downloadable datasets for 34 high-burden US metropolitan areas (2023 reporting year; downloaded October 2025).

S11.2 Mixture-model VL derivation

$$\mu_{\text{mix}} = f_s \mu_{\text{suppressed}} + (1 - f_s) \mu_{\text{unsuppressed}} \quad (\text{S6})$$

Suppressed: $\log_{10}$ mean = 1.5, SD = 0.4. Unsuppressed: $\log_{10}$ mean = 4.5, SD = 1.1. IDU correction: f_s reduced by 0.12 percentage points per percent IDU prevalence, capped at 12 percentage points total.

S11.3 Structural delay: composite indicator (not proxy)

$$\Delta t_{\text{structural}} = \beta_1 (\text{LateDx\%} - 10\%) + \beta_2 \max(90\% - \text{Linkage\%}, 0) + \Delta t_{\text{SDOH}} \quad (\text{S7})$$

$\beta_1 = 1.2$ h/%; $\beta_2 = 0.3$ h/%. SDOH modifier 0–6 h from Gini + poverty. Barrier categories: low (<8 h), moderate (8–18 h), high (18–30 h), severe (≥ 30 h).

Methodological note. City-level structural delay $\Delta t_{\text{structural}}$ is a composite indicator constructed as a calibrated linear function of two AIDSVu surveillance variables (late-diagnosis percentage; linkage-to-care percentage) plus an SDOH modifier. The composite is not a proxy for an unmeasured latent construct under proximal causal inference[12], and the analysis does not invoke the identification conditions required for proxy-based estimation[13]. Coefficients β_1, β_2 are calibrated to reproduce the median city-level access friction; the composite is interpreted as a description of healthcare access friction at the city level, not as an estimand for a separate latent variable.

S11.4 Counterfactual analysis

Parallel simulation assigning all 34 cities a uniform 24-hour access delay isolates structural barrier contribution from community VL. 33/34 cities show observed efficacy > counterfactual (structural delays <24 h). Hartford CT is the sole exception (delay 24.4 h; high-barrier category, 18–30 h). Counterfactual range across cities: 76.0%–82.1%.

S12 Figure Inventory

S12.1 Figure S1 — Focused City Comparison

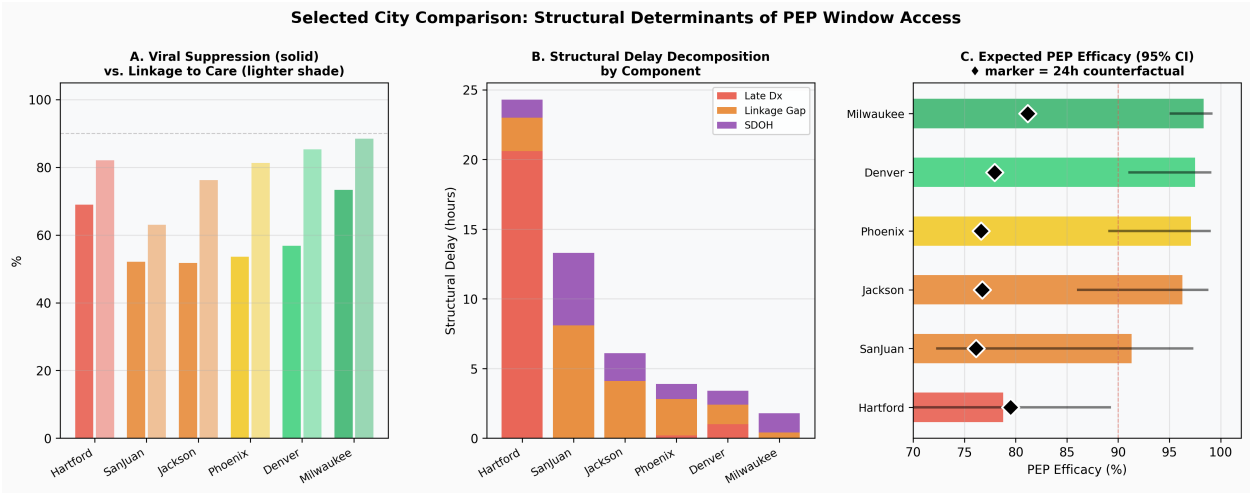
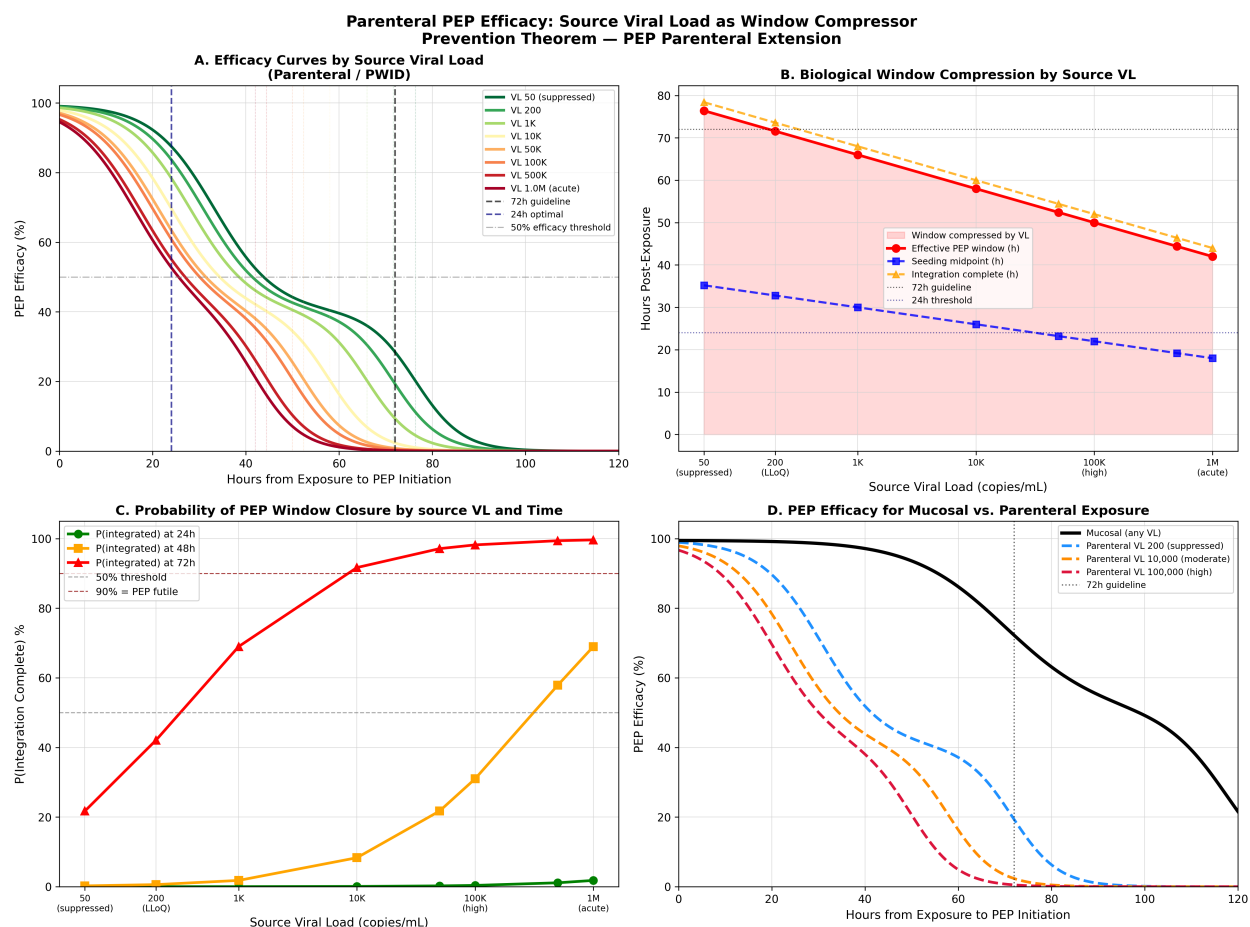


Figure S1: Supplementary Figure S1. Selected city comparison: structural determinants of PEP window access. Six high-vulnerability US cities (Hartford CT, San Juan PR, Jackson MS, Phoenix AZ, Denver CO, Milwaukee WI), ranked left-to-right from highest to lowest structural barrier. (A) Viral suppression rate (solid bars) vs. linkage-to-care rate (lighter shade), per AIDSvU 2023. (B) Structural-delay decomposition by component: late-diagnosis (β_1 contribution), linkage gap (β_2 contribution), and SDOH modifier. Hartford’s 24.4-hour structural delay decomposes as Late Dx $\approx$ 21 h, Linkage Gap $\approx$ 3 h, SDOH $\approx$ 3 h, identifying late-diagnosis as the dominant driver — a Connecticut-specific epidemiological finding consistent with regional surveillance reports. (C) Expected PEP efficacy at city-specific structural delay (95% credible-interval bars; ♦ marker = 24-hour counterfactual). Hartford is the only city among the 34 in Figure 3 where actual expected efficacy falls below the 24-hour counterfactual, reflecting structural-delay overshoot.

S12.2 Figure 1 — Source Viral Load Compression of the Parenteral PEP Window

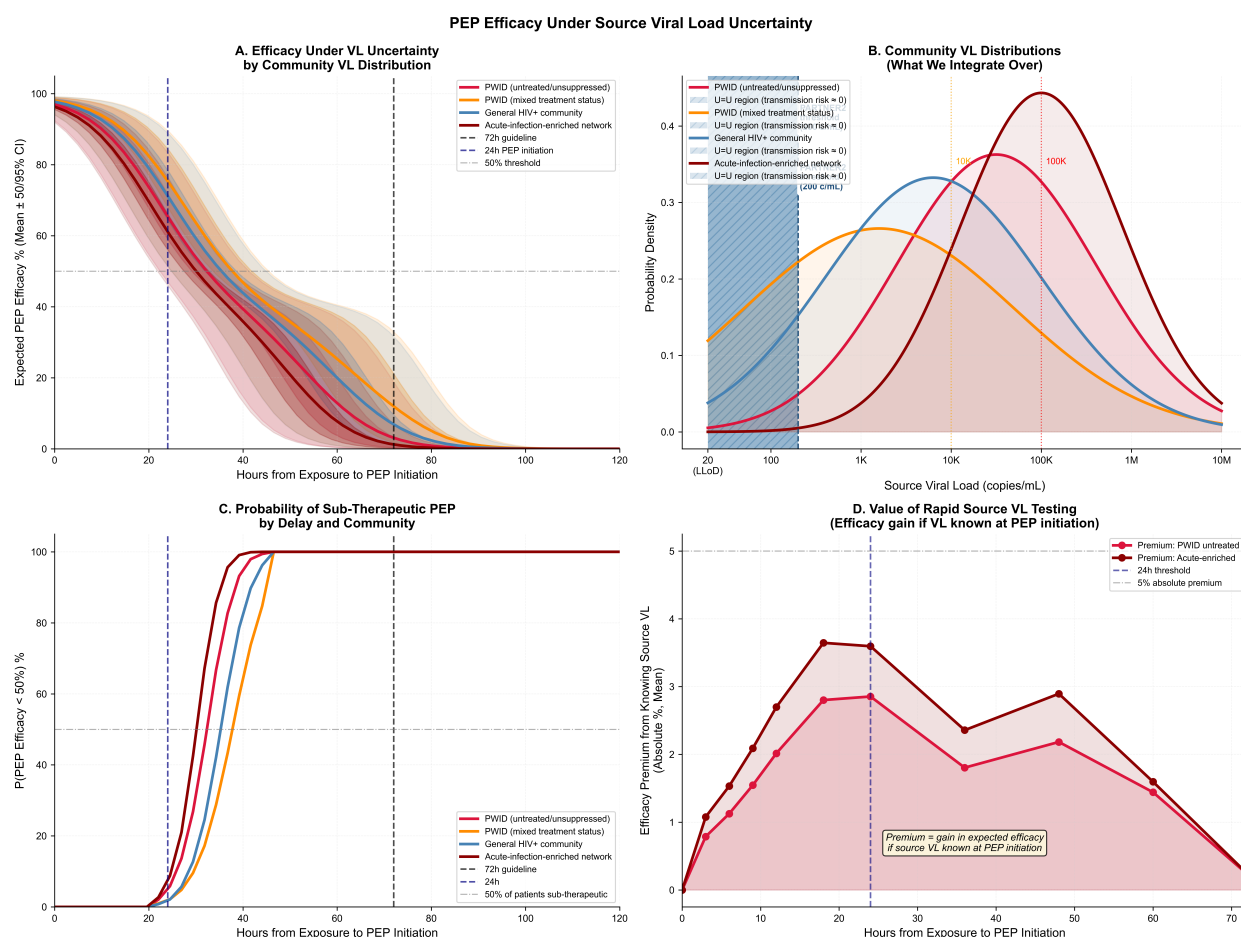
Source: SRC/perelson/parenteral/PEP_parenteral.py (plot_vl_sweep). Output: Figure_1_VL_Compression_PEP



Panel	Title	Description
A	Efficacy curves by source viral load (parenteral / PWID)	$E_{PEP}(t)$ vs. hours post-exposure across 8 source VLs from 50 c/mL (suppressed) through 10^6 c/mL (acute viremia); reference lines at 24h optimal, CDC 72h guideline, and 50% efficacy threshold. Color gradient encodes source VL.
B	Biological window compression by source VL	Effective PEP window, median seeding time, and median integration-complete time as functions of source VL on a logarithmic scale ($10^{1.7}$ – 10^6 c/mL); pink shaded region = compressed actionable window; horizontal references at 24h and 72h.
C	Probability of PEP window closure by source VL and time	$P(\text{integration complete})$ at 24h, 48h, and 72h as functions of source VL; horizontal references at 50% threshold and 90% futility cutoff.
D	PEP efficacy: mucosal vs. parenteral exposure	Mucosal efficacy (post-epithelial attenuation, single curve, all VL) vs. parenteral efficacy stratified by source VL (200, 10,000, 100,000 c/mL); 72h CDC guideline marked.

S12.3 Figure 2 — Stochastic Efficacy Analysis Under Source VL Uncertainty

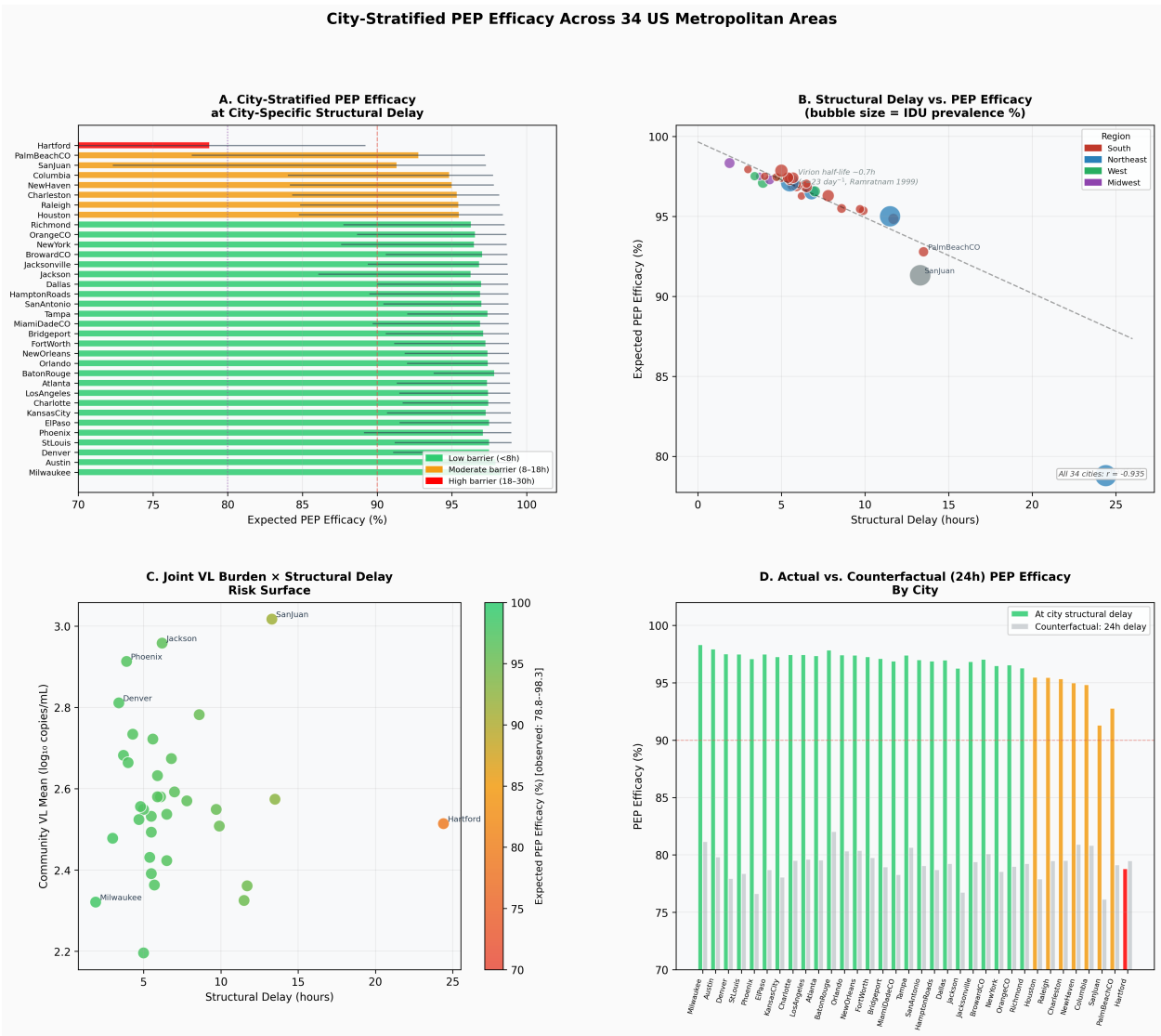
Source: `stochastic_layers/vl_uncertainty.py` (multiscale-derived). Output:



Panel	Title	Description
A	Mean $\pm$ 95% CI Efficacy Curves by Community VL Distribution	Mean $E_{\text{PEP}}(t) \pm 95\%$ CI for all four VL distributions over 0–120 h. Vertical shaded band for U=U scoping. Regime shading overlaid. Primary panel.
B	Community VL Distributions	$\log_{10}$ -normal density plots for all four VL distributions on common x-axis.
C	Probability of Sub-Therapeutic PEP by Delay and Community	$P(\text{efficacy} < 50\%)$ vs. hours post-exposure for all four distributions. Primary panel.
D	VL Knowledge Premium	VL knowledge premium (pp, absolute) vs. hours for all four distributions. Primary panel.

S12.4 Figure 3 — City-Stratified PEP Efficacy Across 34 US Metropolitan Areas

Source: stochastic_layers/city_stratified_figures.py. Outputs:



Panel	Title	Description
A	City-Stratified PEP Efficacy at City-Specific Structural Delay	Ranked horizontal bar chart for 34 high-burden US metropolitan areas (AIDSVu 2023). Bars colored by barrier category: green = low (<8 h), orange = moderate (8–18 h), red = high (>30 h). Range: Milwaukee WI 98.3% to Hartford CT 78.8%. Primary panel; identifies Hartford as the structural-barrier outlier.
B	Structural Delay vs. Expected Efficacy	Scatter plot: x = structural delay (hours), y = expected PEP efficacy (%). Bubble size proportional to IDU prevalence; bubble color by Census region. Pearson $r = -0.935$ annotated; near-deterministic inverse relationship. Labeled outliers: Hartford CT(24.4 h, 23.4% IDU), San Juan PR (13.3 h, 22.2% IDU), Milwaukee WI (1.9 h, lowest barrier). Primary panel.
C	Joint VL Burden $\times$ Structural Delay Risk Surface	2D scatter / heat-map: x = community mean $\log_{10}$ VL, y = structural delay (hours), color = expected PEP efficacy via continuous colormap (%). Shows that structural delay (y) drives the bulk of variance across cities; community VL(x) has limited effect ($r = -0.084$ vs. efficacy).
D	Actual vs. 24-hour Counterfactual PEP Efficacy by City	Paired bars (or lollipop): actual efficacy at city-specific structural delay (filled, color by barrier category) vs. counterfactual at uniform 24-hour delay (open). For 33/34 cities the actual exceeds the counterfactual because current structural delays are shorter than 24 h; Hartford is the sole exception (-0.59 pp deficit). The 19.5 pp range across cities is almost entirely attributable to structural delay ($r = -0.935$) rather than community VL ($r = -0.084$). Primary panel.

Supplementary References

- [1] M.R. Tanner *et al.*, Antiretroviral postexposure prophylaxis — CDC recommendations 2025. *MMWR Recomm. Rep.* (2025).
- [2] J.D. Siliciano, R.F. Siliciano, Enhanced culture assay for detection and quantitation of latently infected, resting CD4⁺ T-cells carrying replication-competent virus in HIV-1-infected individuals, in *Human Retrovirus Protocols* (Humana Press, New Jersey, 2005), vol. 304, pp. 3–16.
- [3] T.W. Chun, D. Finzi, J. Margolick, K. Chadwick, D. Schwartz, R.F. Siliciano, In vivo fate of HIV-1-infected T cells: quantitative analysis of the transition to stable latency. *Nat. Med.* **1**, 1284–1290 (1995).
- [4] A.S. Perelson *et al.*, Decay characteristics of HIV-1-infected compartments during combination therapy. *Nature* **387**, 188–191 (1997).
- [5] A.S. Perelson, A.U. Neumann, M. Markowitz, J.M. Leonard, D.D. Ho, HIV-1 dynamics *in vivo*: virion clearance rate, infected cell life-span, and viral generation time. *Science* **271**, 1582–1586 (1996).
- [6] A. J. McMichael, P. Borrow, G. D. Tomaras, N. Goonetilleke, B. F. Haynes, The immune response during acute HIV-1 infection: clues for vaccine development. *Nat Rev Immunol* **10**, 11–23 (2010).
- [7] H.L. Smith, *Monotone Dynamical Systems* (AMS Mathematical Surveys vol. 41, 1995).
- [8] C.C. Tsai *et al.*, Prevention of SIV infection in macaques by (R)-9-(2-phosphonylmethoxypropyl)adenine. *Science* **270**, 1197–1199 (1995). [Timing data: *J. Virol.* **72**, 4265–4273 (1998)]
- [9] R. A. Otten, D. K. Smith, D. R. Adams, J. K. Pullium, E. Jackson, C. N. Kim, H. Jaffe, R. Janssen, S. Butera, T. M. Folks, Efficacy of postexposure prophylaxis after intravaginal exposure of pig-tailed macaques to a human-derived retrovirus (human immunodeficiency virus type 2). *J Virol* **74**, 9771–9775 (2000).
- [10] J.L. Taylor, A.Y. Walley, A.R. Bazzi, Stuck in the window with you: HIV exposure prophylaxis in the highest risk people who inject drugs. *Subst. Abus.* **40**, 441–443 (2019).
- [11] A.R. Baugher *et al.*, Are we ending the HIV epidemic among persons who inject drugs? Key findings from 19 US cities. *AIDS* **39**(12), 1813–1819 (2025).
- [12] E.J. Tchetgen Tchetgen, A. Ying, Y. Cui, X. Shi, W. Miao, An introduction to proximal causal inference. *Statistical Science* **39**, 375–390 (2024).
- [13] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. (Cambridge University Press, 2009).